Welcome to NexusMaxEnt

- **Maximum Entropy Method (MaxEnt)** is an exceptionally robust mathematical framework for DOSY analysis of mixtures and polydisperse systems (such as synthetic polymers), yet accessible software solutions have historically been scarce.
- **NexusMaxEnt** was developed as a specialized, high-performance platform to perform high-resolution 2D-DOSY Inverse Laplace Transform (ILT) processing via an intuitive GUI.
- The standard engine incorporates three distinct optimization algorithms across CPU and CUDA GPU architectures (**6 solvers in total**): **ADMM KL Mirror Descent**, **Proximal KL Mirror Descent**, and **ADMM LambertW**.
- To foster broader adoption of MaxEnt DOSY analysis, the **ADMM KL Mirror Descent CPU solver (Community Free Edition) is distributed free of charge** (licensed for both academic and commercial use).
- Extended algorithmic suites and ultra-fast CUDA GPU acceleration are available in the **Professional Edition**.

Note: This software and documentation are provided "AS IS" for evaluation and scientific research. Individualized technical support or custom modifications are not accommodated. Please utilize GitHub Discussions for community exchange and bug reporting.

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Installing NexusMaxEnt

- **Zero-Configuration Standalone Package**:
  - NexusMaxEnt is engineered on a unified Python 3.11 foundation.
  - All necessary runtime binaries and scientific libraries are fully bundled; no separate Python environment or package management is required on your workstation.
- **Running the Installer**:
  - Launch the distributed setup executable (`.exe`) and install the program into your preferred local directory.
- **Launching the Application**:
  - Execute `NexusMaxEnt.exe` located within the destination folder.
  - Creating a desktop shortcut to `NexusMaxEnt.exe` is recommended for operational convenience.

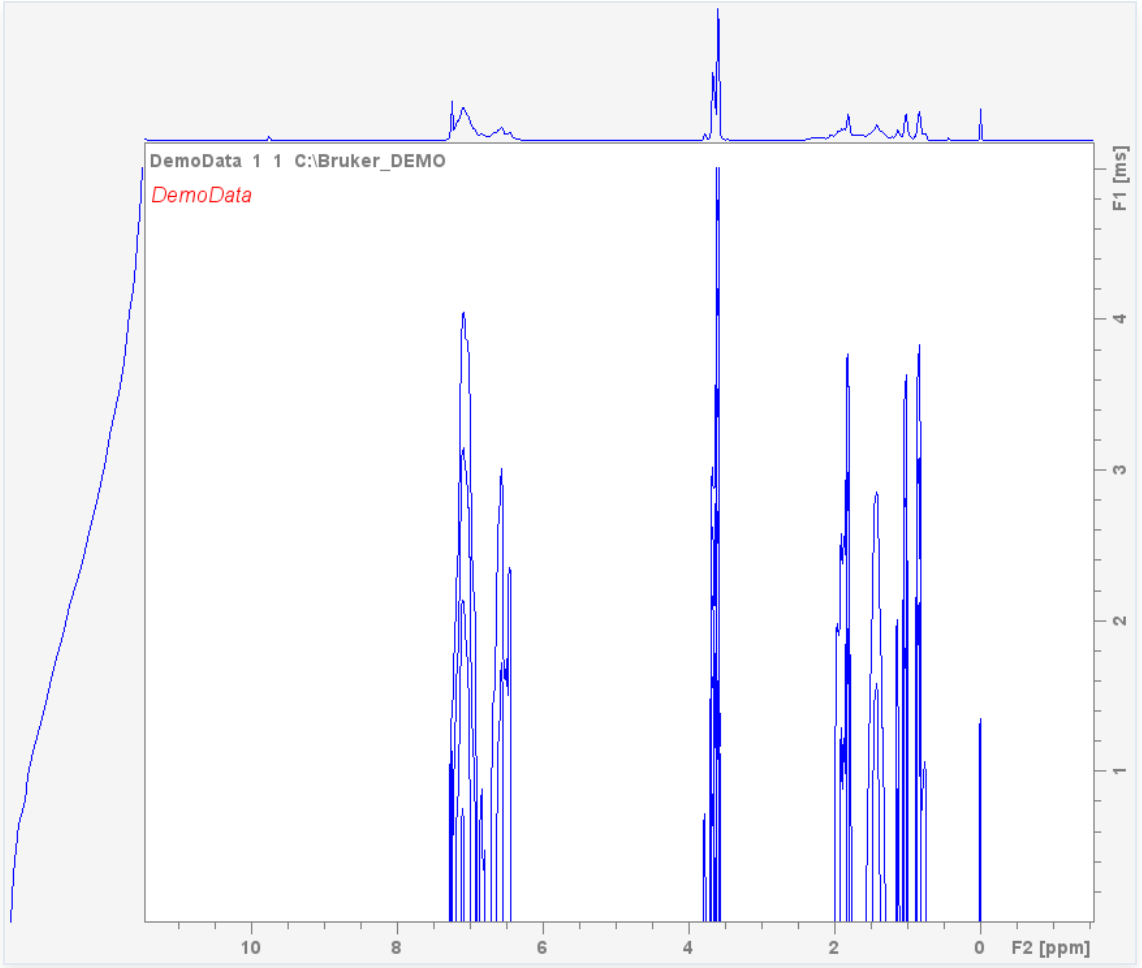
Note: Utilizing the GPU-accelerated engine (Professional Edition) requires an NVIDIA GPU (Compute Capability ≥ 7.5 recommended: GeForce GTX 1650 or newer) along with an up-to-date NVIDIA display driver.

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1. Data Preparation (Bruker)

- Open your 2D DOSY dataset in TopSpin and define the Fourier transform data points using the `si` command:
  - **F2 axis (Chemical Shift)**: Set to `8192`
  - **F1 axis (Gradient Dimension)**: Match the actual number of measured gradient steps (F1 `TD`)
- Set the exponential apodization parameter `lb` to approximately **2.0 Hz** to suppress truncation artifacts (ringing distortion from truncated FIDs).
- Execute `xf2` to perform 1D Fourier transformation along the direct F2 dimension only.
- Accurately correct the F2 phase (zero- and first-order).
- Perform baseline correction across all rows using `abs2`.
- Preprocessing is complete when a consistent stack of 1D spectra with flat baselines is confirmed, as shown on the right.

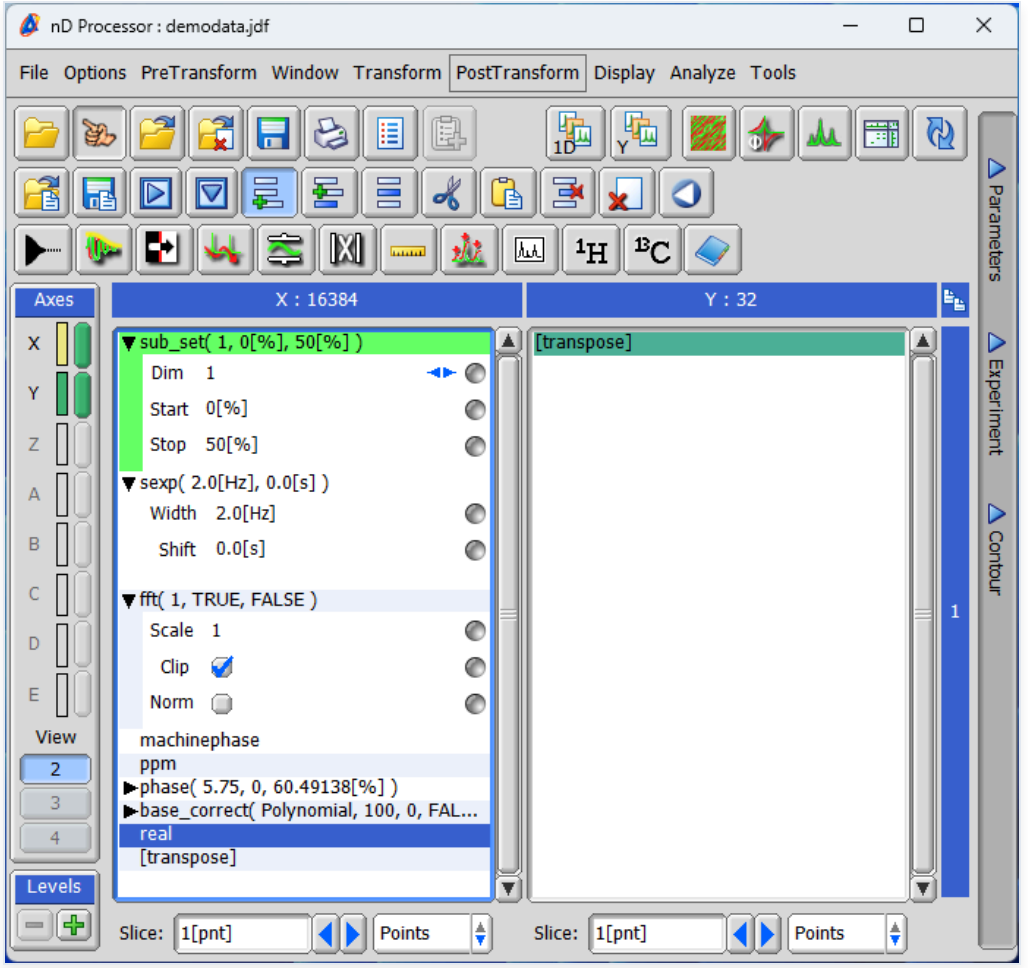
Note: The F2 point size and lb value represent standard recommendations. The solver imposes no hard limits on data array dimensions.



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1. Data Preparation (JEOL - Part 1)

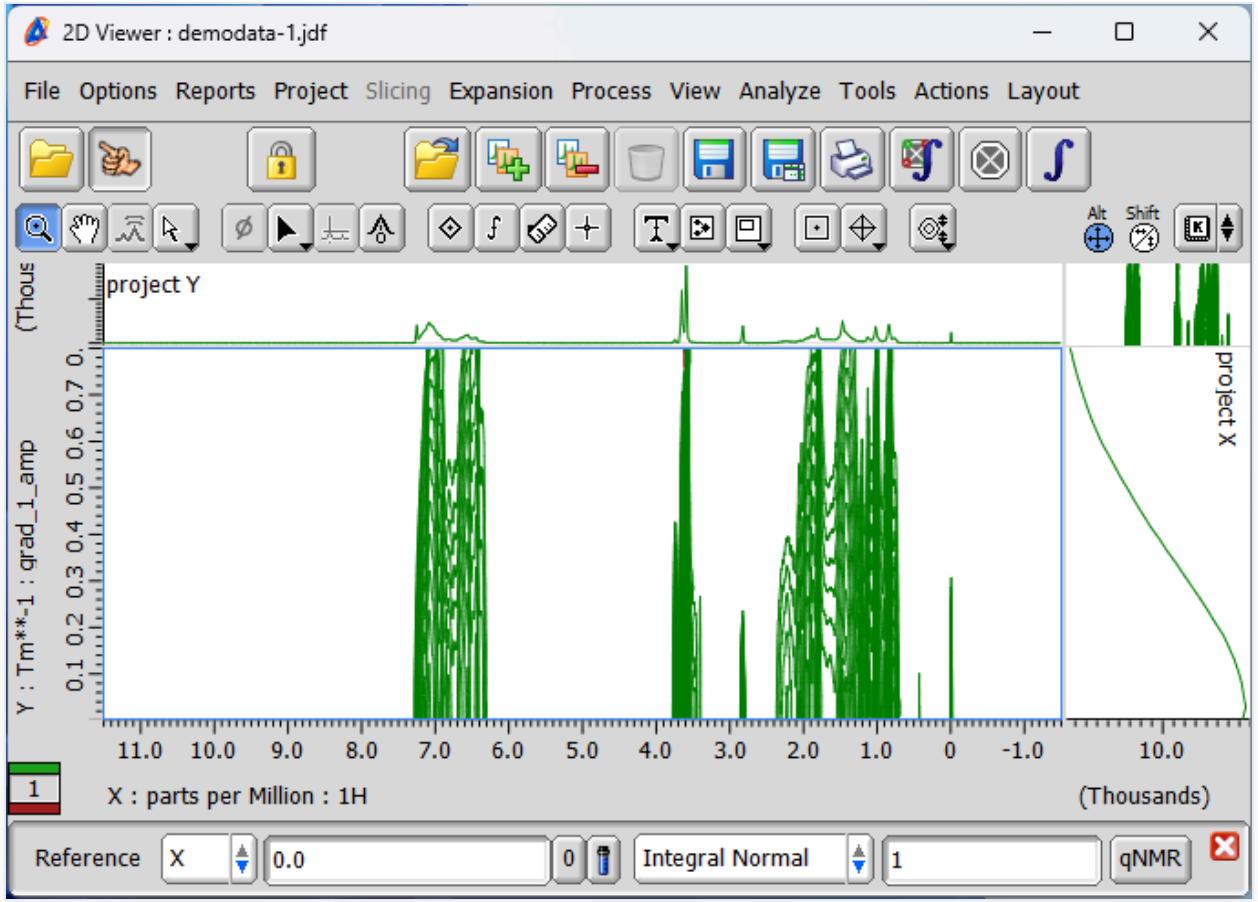
- In DOSY processing, excessive digital resolution along the chemical shift axis increases computational burden without providing physical insight.
- In this workflow, the **Sub Set** function is utilized to truncate the FID data length by half (**Stop 50%**).
- NexusMaxEnt is optimized assuming approximately **8,000 data points** along the chemical shift axis (no intrinsic limits exist; adjust the **Sub Set Stop[%]** based on original acquisition size).
- Apply a single exponential window function (`sexp`) at approximately **2.0 Hz** to mitigate truncation artifacts.
- Execute **FFT** (the Norm setting may be left either ON or OFF).
- Perform phase and baseline corrections, and set the horizontal scale to **ppm**.
- Finally, append `real` to retain only the real component (imaginary components are unnecessary for ILT).



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1. Data Preparation (JEOL - Part 2)

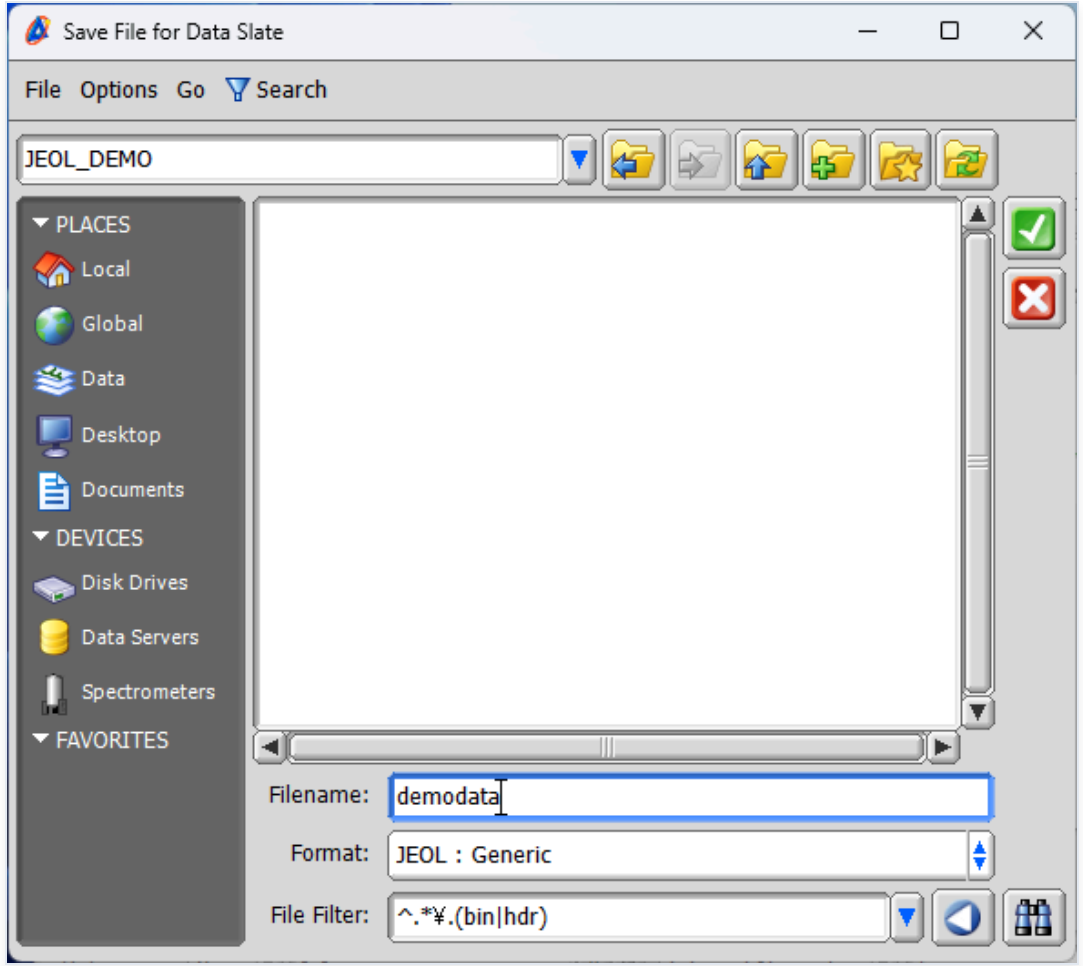
- Open the processed dataset in **2D Viewer**.
- Confirm that the horizontal axis represents chemical shift (ppm) and the vertical axis represents PFG magnetic field gradient amplitude ($\mathrm{T/m}$ or $\mathrm{mT/m}$).
- Preprocessing is verified when a coherent spectral series exhibiting monotonic signal attenuation with increasing gradient strength is observed, as illustrated on the right.



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1. Data Preparation (JEOL - Part 3)

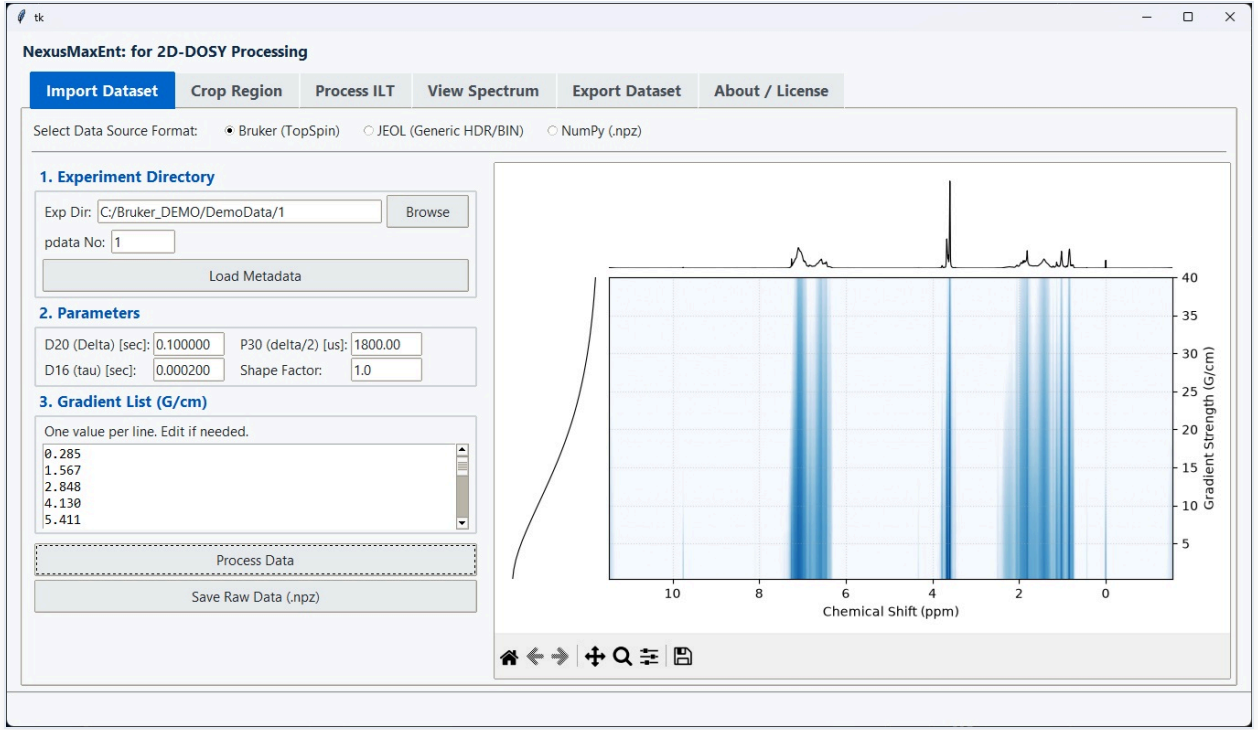
- In the 2D Viewer or Data Slate menu, navigate to **File > Save File**.
- Select `JEOL : Generic` as the export format and execute the save operation.
- The following paired files will be generated:
  - `.hdr` : Header metadata (spectrometer frequency, gradient list, and acquisition parameters)
  - `.bin` : Pure binary numerical matrix containing the real 2D spectrum
- NexusMaxEnt natively parses this Generic pair (`.hdr` / `.bin`) for ILT inversion.



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2. Import Dataset (Bruker)

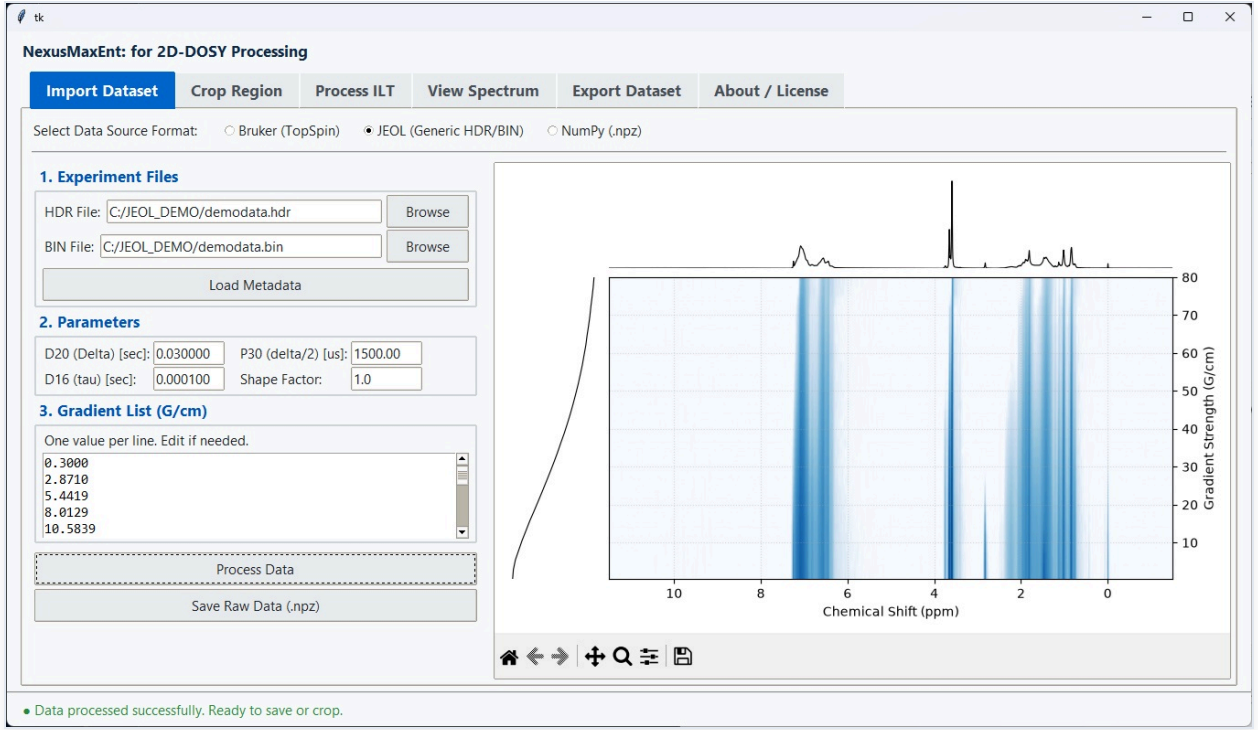
- **Experiment Directory**: Specify the directory containing `difflist` and the raw `ser` (or processed `2rr`) files.
- **pdata No.**: Enter the processed sub-directory number (e.g., `1`) and click **Load Metadata**.
- Verify that **D20** (Δ: diffusion delay), **P30** ($\delta/2$: gradient pulse duration), and **D16** (τ: eddy current delay) are populated accurately.
  - *Calibrated for bipolar pulse pair (BPP) sequences (e.g., ledbpgp2s). For standard STE sequences, enter effective $\delta/2$ into P30, and set D16 to 0.*
- **Shape Factor**:
  - When acquired and processed using TopSpin's `xau dosy`, gradient shape factors are already folded into `difflist`; set **Shape Factor** to `1.0`.
- Click **Process Data** to import (optionally archive via **Save Raw Data (.npz)**).



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2. Import Dataset (JEOL)

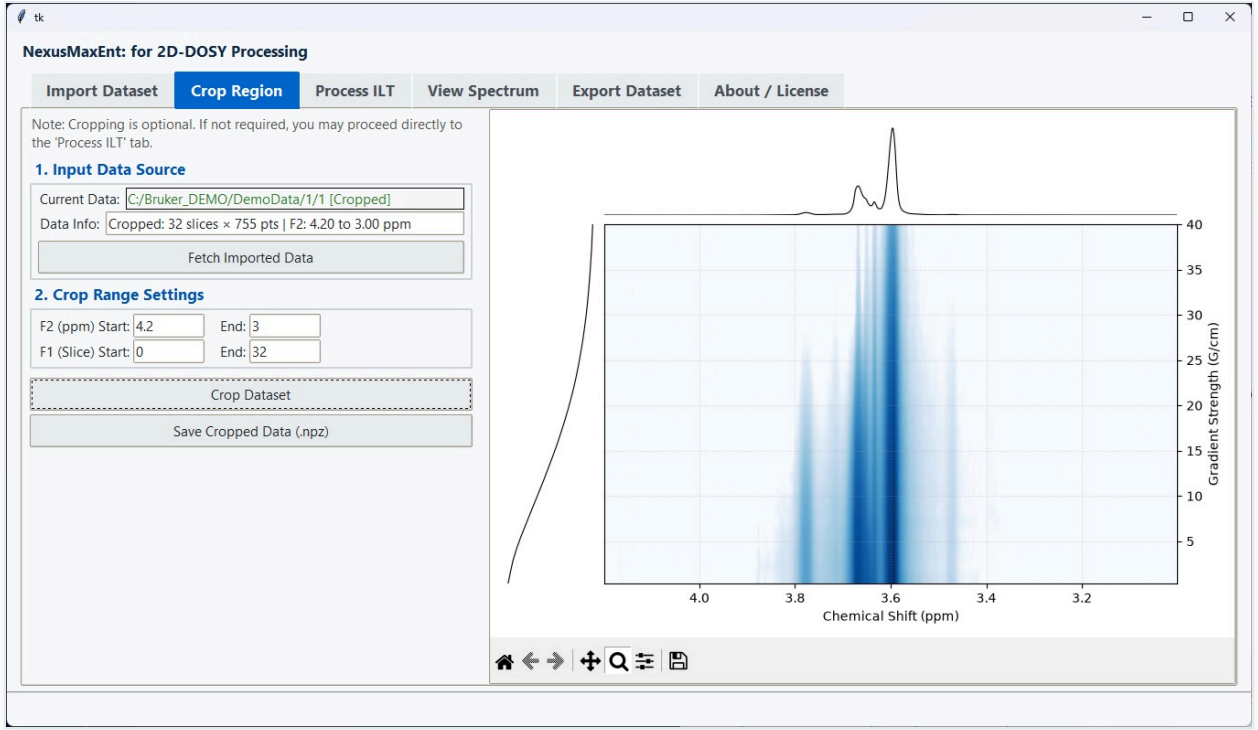
- Specify the path to the JEOL Generic file and click **Load Metadata** to parse experimental parameters.
  - *Verify that diffusion time units (seconds vs. milliseconds) are interpreted correctly; adjust manually if necessary.*
- **Shape Factor**:
  - Enter the effective gradient integral factor corresponding to the applied PFG waveform.
  - SQUARE: `1.0`
  - SINE: `0.6366` (theoretical standard; please refer to probe calibration documentation)
- After confirming parameters, click **Process Data**.
- Click **Save Raw Data (.npz)** to cache the imported dataset.



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3. Crop Region

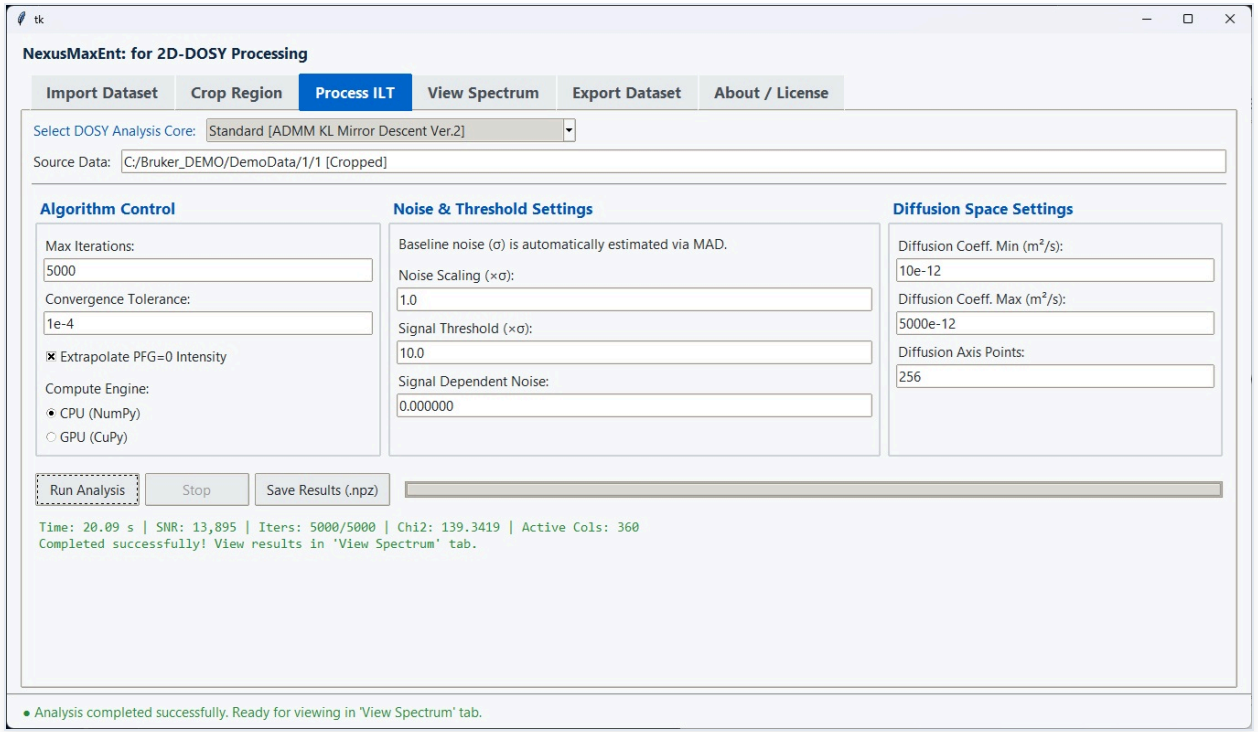
- Click **Fetch Imported Data** to transfer the preprocessed matrix into the cropping workspace.
- To reduce execution time and focus on target resonance peaks, define the analysis boundaries (F1 slice indices, F2 ppm range) and click **Crop Dataset**.
- If full-spectrum processing is desired without cropping, skip this step and proceed directly to **Process ILT**.
- Cropped datasets can be saved for future batch analysis using **Save Cropped Data (.npz)**.



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4. Process ILT

- The Community Free Edition includes the robust and high-precision **ADMM KL Mirror Descent Ver.2** (CPU solver), which operates optimally with default parameters.
- **Convergence Tolerance**: Termination threshold based on the relative spectral change between iterations.
- **Noise Scaling (×σ)**: Scaling factor applied to baseline noise automatically estimated via Median Absolute Deviation (MAD) (Default: `1.0`).
- **Signal Threshold (×σ)**: Threshold factor relative to estimated noise below which spectral columns are bypassed (Default: $\geq$ `10.0` recommended).
- **Signal Dependent Noise**: Accounts for instrument amplitude instability and phase jitter (Default: `0` ; set to ~ `0.001` [0.1%] if overfitting artifacts appear).

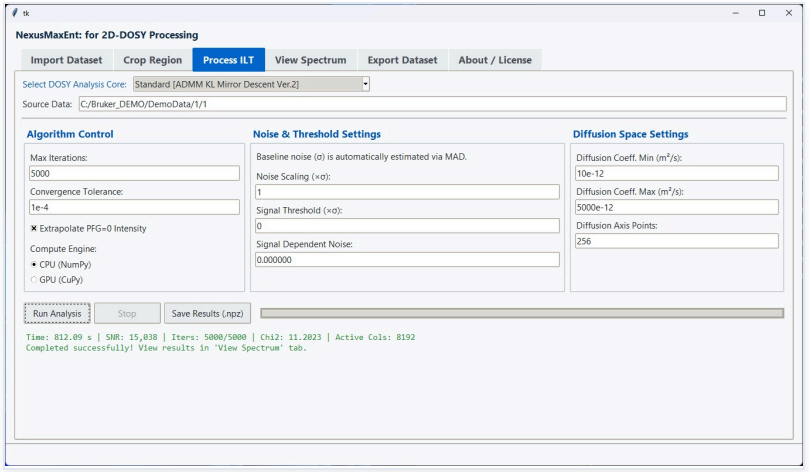


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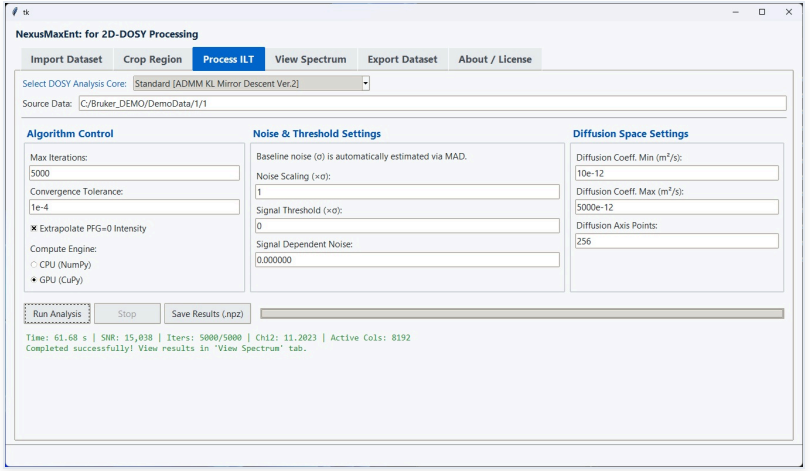
4. Process ILT (Performance Comparison)

- **CPU Processing (NumPy / Community Free Edition)**:
  - Analyzing all 8,192 channels (Signal Threshold = 0) completes in **~812 s** (benchmarked on AMD Ryzen 5 3600 / 6-Core CPU).
  - Cropping regions of interest or applying realistic signal thresholds reduces runtime to tens of seconds.
- **GPU Acceleration (CuPy / Professional Edition)**:
  - Under identical conditions, runtime drops to **~62 s** (**~13× acceleration**, benchmarked on NVIDIA RTX 3060).
  - Even entry-level GPUs (e.g., GTX 1650) deliver dramatic speedups over high-end multi-core CPUs.

CPU (NumPy) : 812.09 s



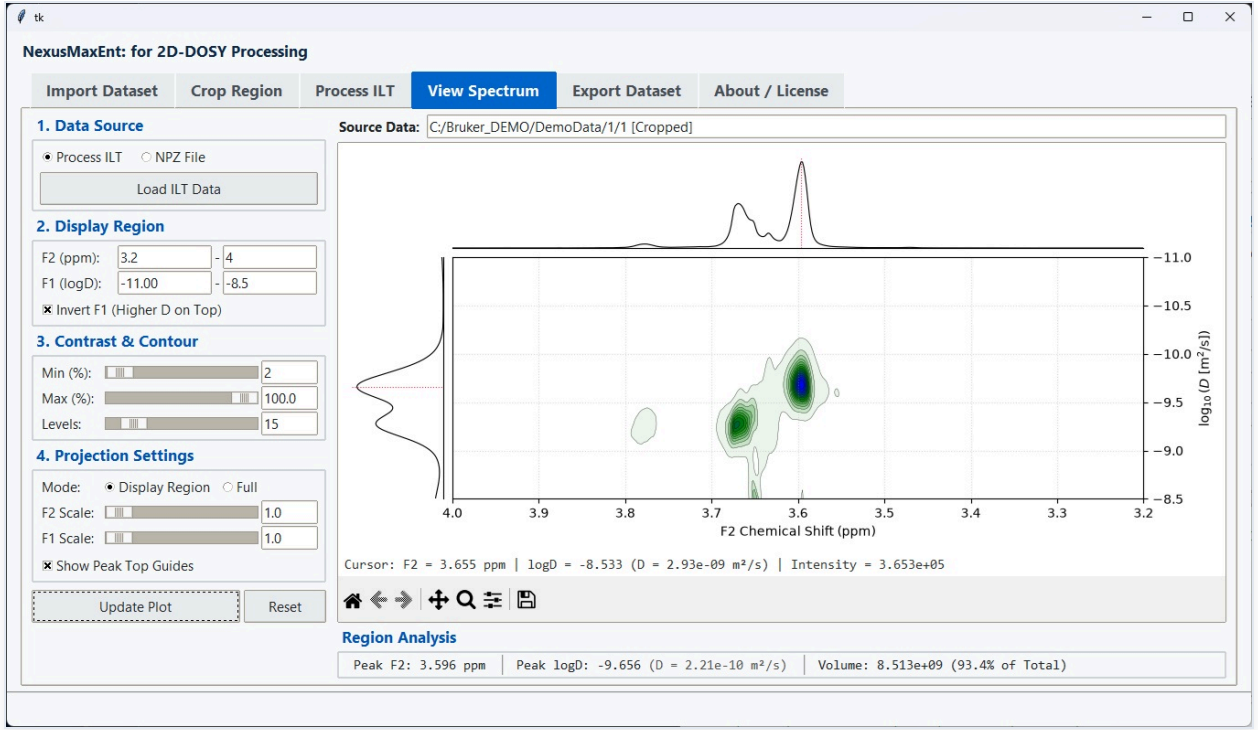
GPU (CuPy) : 61.68 s (~13× Faster)



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5. View Spectrum (Part 1)

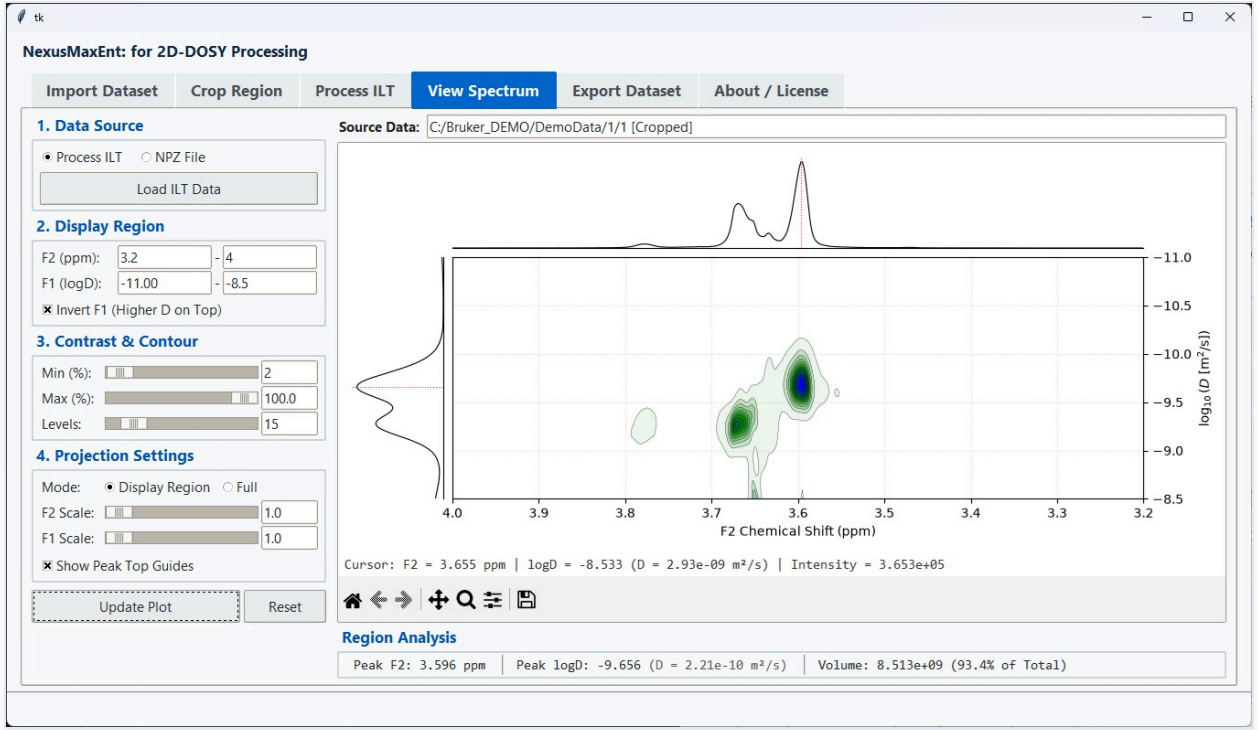
- Click **Load ILT Data** or open a processed `.npz` file to render the resolved 2D-DOSY spectrum.
- The integrated Matplotlib interface provides real-time cursor tracking of chemical shifts (ppm) and diffusion coefficients ($\log_{10} D$ / D), alongside interactive zoom and pan controls.
- **Contrast & Contour**: Dynamically adjust the display floor (**Min**), ceiling (**Max**), and contour resolution (**Levels**).
- **Projection Settings**: Adjust display gains for the top projection (1D NMR) and side projection (diffusion coefficient distribution).
- The underlying `.npz` structure uses an open format, facilitating custom high-resolution figure generation using external Python scripts.



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5. View Spectrum (Part 2)

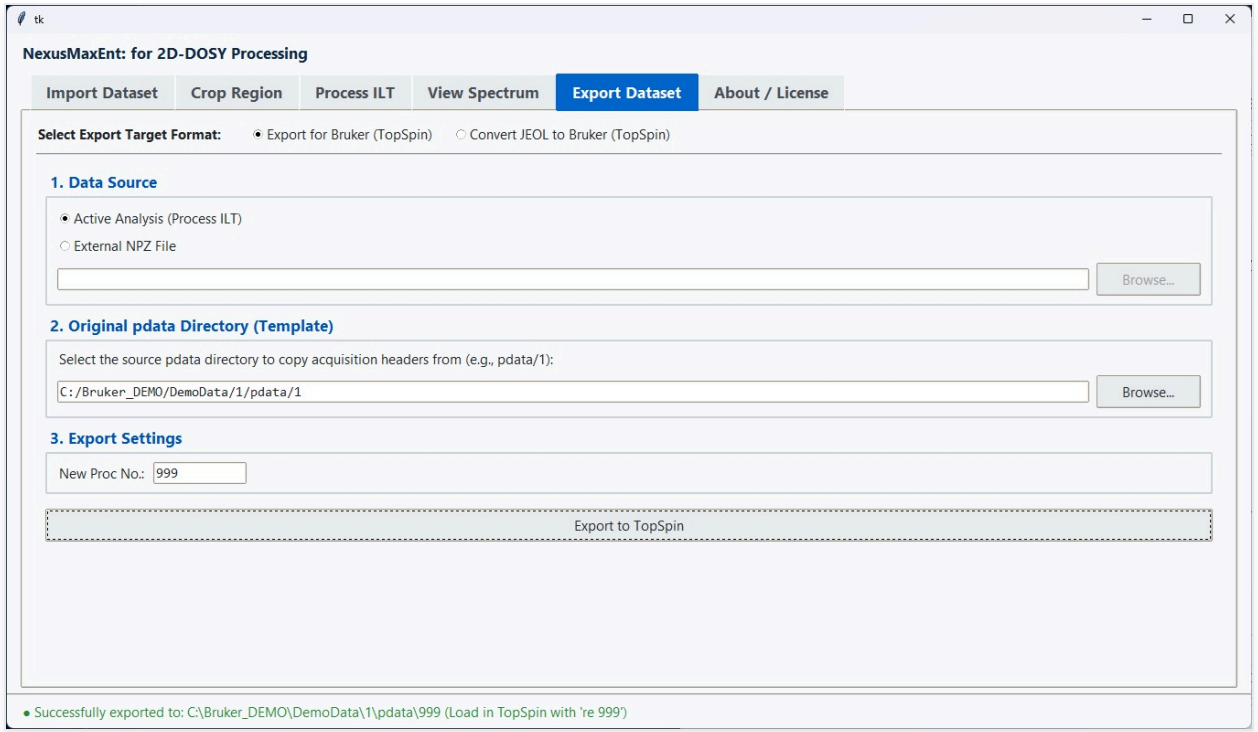
- Specifying numerical boundaries under **2. Display Region** automatically computes quantitative parameters within the **Region Analysis** panel:
  - **Peak F2**: Peak top chemical shift of the integrated regional projection
  - **Peak logD**: Diffusion coefficient apex (with equivalent physical D value)
  - **Volume**: Integrated regional volume and its relative proportion (% of total)
- *Note: Manual zooming via the toolbar magnifying glass does not update Region Analysis. Use **Display Region** for quantitative evaluation.*
- Toggling **Show Peak Top Guides** overlays red dashed fiducial lines tracking projection apexes.



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6. Export Dataset (Bruker)

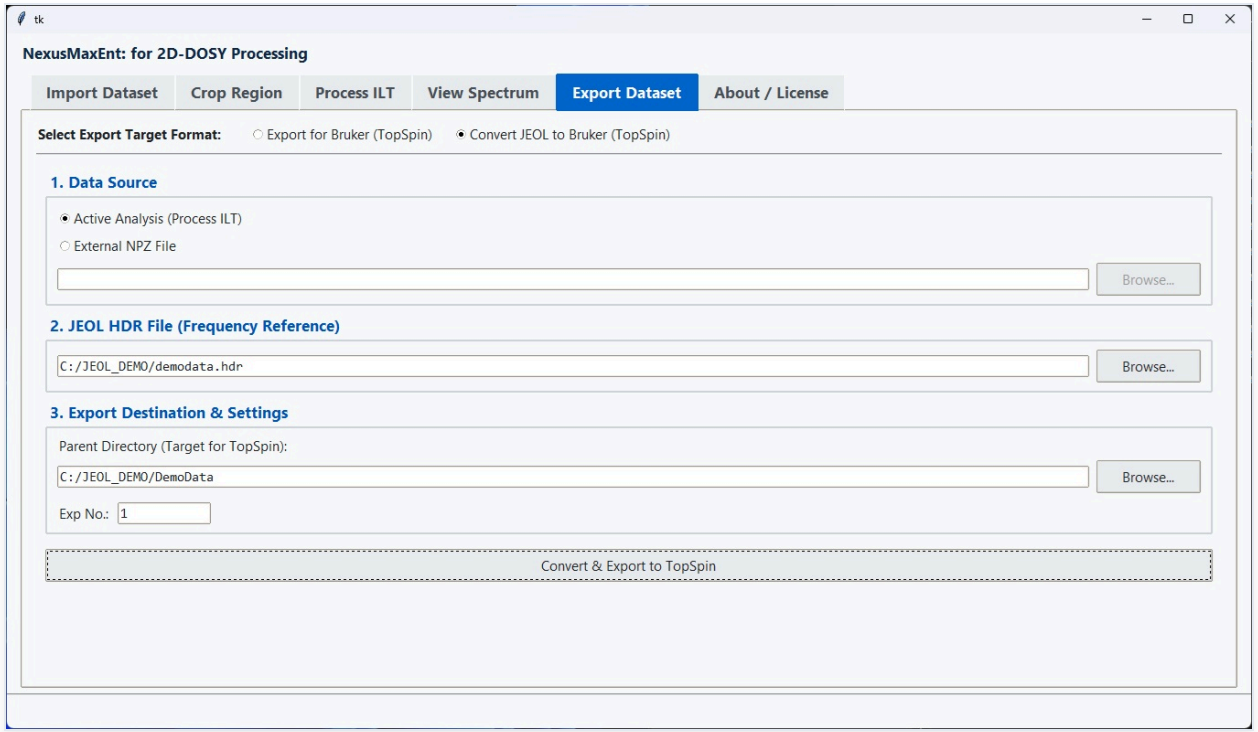
- Seamlessly export ILT-resolved datasets into native Bruker TopSpin formats directly from memory or stored `.npz` archives.
- NexusMaxEnt preserves acquisition headers from the template `pdata` while synthesizing the newly calculated diffusion axis and data points into a **new pdata directory**.
- **Original pdata Directory**: Select the source template directory (e.g., `pdata/1`).
- **New Proc No.**: Specify an unused process number (e.g., `999`) and click **Export to TopSpin**.
- Open the dataset in TopSpin (or execute `re 999`) to immediately inspect, integrate, and plot the 2D DOSY map natively.



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6. Export Dataset (JEOL)

- Converts and exports JEOL Generic datasets into a **TopSpin-compatible directory hierarchy**.
- **JEOL HDR File**: Specify the associated `.hdr` file to inherit spectrometer frequency and metadata correctly (typically loaded during import).
- **Parent Directory**: Specify the destination root directory.
- **Exp No.**: Assign an experiment number; TopSpin-compliant directory structures (`pdata/1/2rr` , etc.) are synthesized automatically.
- The exported dataset can be opened directly within TopSpin for unified cross-vendor data management and direct comparison.



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7. About & License

- **Licensing & Scope of Use (Community Free Edition)**:
  - The bundled ADMM Mirror CPU solver component is provided **free of charge for academic, personal, and commercial purposes**.
  - Direct commercial resale, bundling into commercial distributions, and reverse engineering are strictly prohibited.
- **Academic & Professional Citation**:
  - When publishing research utilizing results from NexusMaxEnt, formal citation of the author and software package is kindly requested:
  - `Tanigawa, R. (2026). NexusMaxEnt (Version 1.4.0) [Computer software]. GitHub: https://github.com/Ryuichi-Tanigawa/NexusMaxEnt`
- **Community Forum & Support Policy**:
  - Individualized private inquiries and email consultations are not accommodated.
  - Questions, bug reports, and workflow discussions are warmly welcomed on **GitHub Discussions**.
- **Disclaimer**:
  - This software is provided "AS IS" without warranty of any kind. Please refer to the bundled `LICENSE.txt` for detailed legal terms.

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